GATE 2009 - Electronics and Communication Engineering

Note: the supplied question paper has no answer key and this scanned paper has not yet been solved, so only the questions are given.

GATE 2009 | Q.01

Q.1 The order of the differential equation $\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^3 + y^4 = e^{-t}$ is

(A) 1 (B) 2 (C) 3 (D) 4

Answer: not available

GATE 2009 | Q.02

Q.2 The Fourier series of a real periodic function has only

P. cosine terms if it is even
Q. sine terms if it is even
R. cosine terms if it is odd
S. sine terms if it is odd

Which of the above statements are correct ?

(A) P and S (B) P and R (C) Q and S (D) Q and R

Answer: not available

GATE 2009 | Q.03

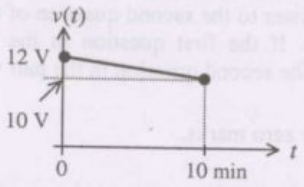
Q.3 A function is given by $f(t) = \sin^2 t + \cos 2t$. Which of the following is true ?

(A) f has frequency components at 0 and $1/2\pi$ Hz
(B) f has frequency components at 0 and $1/\pi$ Hz
(C) f has frequency components at $1/2\pi$ and $1/\pi$ Hz
(D) f has frequency components at 0, $1/2\pi$ and $1/\pi$ Hz

Answer: not available

GATE 2009 | Q.04

Q.4 A fully charged mobile phone with a 12 V battery is good for a 10 minute talk-time. Assume that, during the talk-time, the battery delivers a constant current of 2 A and its voltage drops linearly from 12 V to 10 V as shown in the figure. How much energy does the battery deliver during this talk-time ?



(A) 220 J (B) 12 kJ (C) 13.2 kJ (D) 14.4 kJ

Answer: not available

GATE 2009 | Q.05

Q.5 In an **n**-type silicon crystal at room temperature, which of the following can have a concentration of $4 \times 10^{19}\ cm^{-3}$?

(A) Silicon atoms (B) Holes (C) Dopant atoms (D) Valence electrons

Answer: not available

GATE 2009 | Q.06

Q.6 The full forms of the abbreviations TTL and CMOS in reference to logic families are

(A) Triple Transistor Logic and Chip Metal Oxide Semiconductor
(B) Tristate Transistor Logic and Chip Metal Oxide Semiconductor
(C) Transistor Transistor Logic and Complementary Metal Oxide Semiconductor
(D) Tristate Transistor Logic and Complementary Metal Oxide Silicon

Answer: not available

GATE 2009 | Q.07

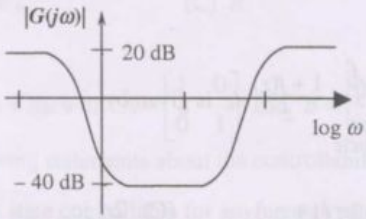
Q.7 The ROC of Z-transform of the discrete time sequence $x(n) = \left(\frac{1}{3}\right) u(n) - \left(\frac{1}{2}\right) u(-n-1)$ is

(A) $\frac{1}{3} < |z| < \frac{1}{2}$ (B) $|z| > \frac{1}{2}$ (C) $|z| < \frac{1}{3}$ (D) $2 < |z| < 3$

Answer: not available

GATE 2009 | Q.08

Q.8 The magnitude plot of a rational transfer function $G(s)$ with real coefficients is shown below. Which of the following compensators has such a magnitude plot ?

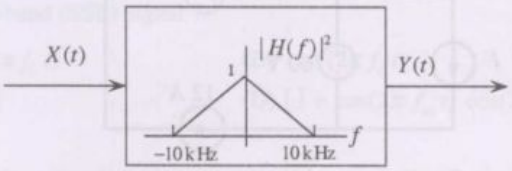


(A) Lead compensator (B) Lag compensator
(C) PID compensator (D) Lead-lag compensator

Answer: not available

GATE 2009 | Q.09

Q.9 A white noise process $X(t)$ with two-sided power spectral density 1×10^{-10} W/Hz is input to a filter whose magnitude squared response is shown below.



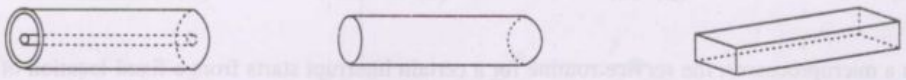
The power of the output process $Y(t)$ is given by

(A) 5×10^{-7} W (B) 1×10^{-6} W (C) 2×10^{-6} W (D) 1×10^{-5} W

Answer: not available

GATE 2009 | Q.10

Q.10 Which of the following statements is true regarding the fundamental mode of the metallic waveguides shown ?



P: Coaxial Q: Cylindrical R: Rectangular

(A) Only P has no cutoff-frequency
(B) Only Q has no cutoff-frequency
(C) Only R has no cutoff-frequency
(D) All three have cut-off frequencies

Answer: not available

GATE 2009 | Q.11

Q.11 A fair coin is tossed 10 times. What is the probability that ONLY the first two tosses will yield heads ?

(A) $\left(\frac{1}{2}\right)^2$ (B) ${}^{10}C_2\left(\frac{1}{2}\right)^2$ (C) $\left(\frac{1}{2}\right)^{10}$ (D) ${}^{10}C_2\left(\frac{1}{2}\right)^{10}$

EC

Q.12 If the power spectral density of stati...

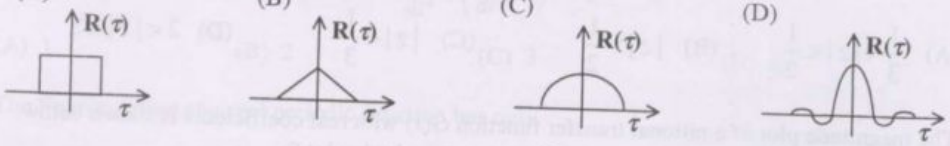
Answer: not available

GATE 2009 | Q.12

EC

Q.12 If the power spectral density of stationary random process is a sinc-squared function of frequency, the shape of its autocorrelation is

(A) (B) (C) (D)



Answer: not available

GATE 2009 | Q.13

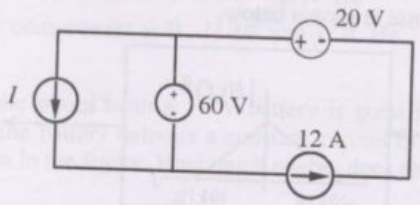
Q.13 If $f(z) = c_0 + c_1 z^{-1}$, then $\oint_{\text{unit circle}} \frac{1+f(z)}{z} dz$ is given by

(A) $2\pi c_1$ (B) $2\pi(1+c_0)$ (C) $2\pi j c_1$ (D) $2\pi j(1+c_0)$

Answer: not available

GATE 2009 | Q.14

Q.14 In the interconnection of ideal sources shown in the figure, it is known that the 60 V source is absorbing power.



Which of the following can be the value of the current source I ?

(A) 10 A (B) 13 A (C) 15 A (D) 18 A

Answer: not available

GATE 2009 | Q.15

Q.15 The ratio of the mobility to the diffusion coefficient in a semiconductor has the units

(A) V^{-1} (B) $cm \cdot V^{-1}$ (C) $V \cdot cm^{-1}$ (D) $V \cdot s$

Answer: not available

GATE 2009 | Q.16

Q.16 In a microprocessor, the service routine for a certain interrupt starts from a fixed location of memory which cannot be externally set, but the interrupt can be delayed or rejected. Such an interrupt is

(A) non-maskable and non-vectored

(B) maskable and non-vectored

(C) non-maskable and vectored

(D) maskable and vectored

EC

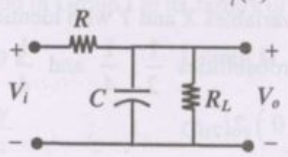
4/24

Q.17 If the transfer function of the following network is $\frac{V_o(s)}{\quad} = \frac{1}{\quad}$

Answer: not available

GATE 2009 | Q.17

Q.17 If the transfer function of the following network is $\frac{V_o(s)}{V_i(s)} = \frac{1}{2+sCR}$,



the value of the load resistance R_L is

(A) $R/4$ (B) $R/2$ (C) R (D) $2R$

Answer: not available

GATE 2009 | Q.18

Q.18 Consider the system $\frac{dx}{dt} = Ax + Bu$ with $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} p \\ q \end{bmatrix}$ where p and q are arbitrary real numbers. Which of the following statements about the controllability of the system is true ?

(A) The system is completely state controllable for any nonzero values of p and q
(B) Only $p = 0$ and $q = 0$ result in controllability
(C) The system is uncontrollable for all values of p and q
(D) We cannot conclude about controllability from the given data

Answer: not available

GATE 2009 | Q.19

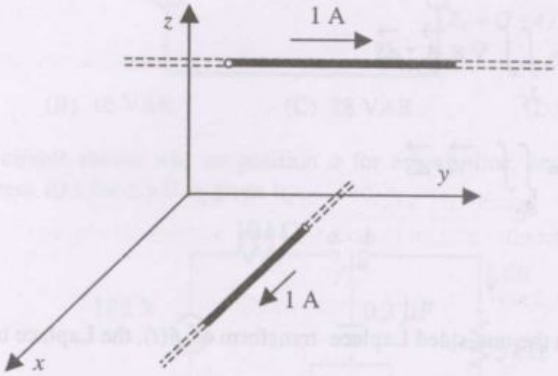
Q.19 For a message signal $m(t) = \cos(2\pi f_m t)$ and carrier of frequency f_c, which of the following represents a single-side-band (SSB) signal ?

(A) $\cos(2\pi f_m t)\cos(2\pi f_c t)$ (B) $\cos(2\pi f_c t)$
(C) $\cos[2\pi(f_c + f_m)t]$ (D) $[1 + \cos(2\pi f_m t)]\cos(2\pi f_c t)$

Answer: not available

GATE 2009 | Q.20

Q.20 Two infinitely long wires carrying current are as shown in the figure below. One wire is in the y-z plane and parallel to the y-axis. The other wire is in the x-y plane and parallel to the x-axis. Which components of the resulting magnetic field are non-zero at the origin ?



(A) x, y, z components
(B) x, y components
(C) y, z components
(D) x, z components

Q. 21 to Q. 60 carry two marks each.

Answer: not available

GATE 2009 | Q.21

Q.21 Consider two independent random variables X and Y with identical distributions. The variables X and Y take values 0, 1 and 2 with probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$ respectively. What is the conditional probability $P\left(X+Y=2 \middle| X-Y=0\right)$?

(A) 0
(B) $\frac{1}{16}$
(C) $\frac{1}{6}$
(D) 1

Answer: not available

GATE 2009 | Q.22

Q.22 The Taylor series expansion of $\frac{\sin x}{x-\pi}$ at $x=\pi$ is given by

(A) $1+\frac{(x-\pi)^2}{3!}+\ldots$
(B) $-1-\frac{(x-\pi)^2}{3!}+\ldots$
(C) $1-\frac{(x-\pi)^2}{3!}+\ldots$
(D) $-1+\frac{(x-\pi)^2}{3!}+\ldots$

Answer: not available

GATE 2009 | Q.23

Q.23 If a vector field $\vec{V}$ is related to another vector field $\vec{A}$ through $\vec{V} = \nabla \times \vec{A}$, which of the following is true ? *Note:* C and S_C refer to any closed contour and any surface whose boundary is C.

(A) $\oint_C \vec{V} \cdot \vec{dl} = \iint_{S_C} \vec{A} \cdot \vec{dS}$

(B) $\oint_C \vec{A} \cdot \vec{dl} = \iint_{S_C} \vec{V} \cdot \vec{dS}$

(C) $\oint_C \nabla \times \vec{V} \cdot \vec{dl} = \iint_{S_C} \nabla \times \vec{A} \cdot \vec{dS}$

(D) $\oint_C \nabla \times \vec{A} \cdot \vec{dl} = \iint_{S_C} \vec{V} \cdot \vec{dS}$

Answer: not available

GATE 2009 | Q.24

Q.24 Given that $F(s)$ is the one-sided Laplace transform of $f(t)$, the Laplace transform of $\int_0^t f(\tau)d\tau$ is

(A) $sF(s) - f(0)$ (B) $\frac{1}{s}F(s)$ (C) $\int_0^s F(\tau)d\tau$ (D) $\frac{1}{s}[F(s) - f(0)]$

Answer: not available

GATE 2009 | Q.25

Q.25 Match each differential equation in Group I to its family of solution curves from Group II.

Group I		Group II	
P.	$\frac{dy}{dx} = \frac{y}{x}$	1.	Circles
Q.	$\frac{dy}{dx} = -\frac{y}{x}$	2.	Straight lines
R.	$\frac{dy}{dx} = \frac{x}{y}$	3.	Hyperbolas
S.	$\frac{dy}{dx} = -\frac{x}{y}$		

(A) P-2, Q-3, R-3, S-1 (B) P-1, Q-3, R-2, S-1 (C) P-2, Q-1, R-3, S-3 (D) P-3, Q-2, R-1, S-2

Answer: not available

GATE 2009 | Q.26

Q.26 The eigen values of the following matrix are

$$\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

(A) $3,\ 3+5j,\ 6-j$

(B) $-6+5j,\ 3+j,\ 3-j$

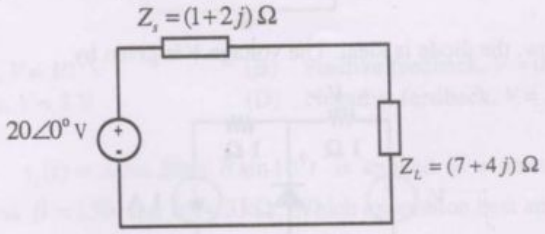
(C) $3+j,\ 3-j,\ 5+j$

(D) $3,\ -1+3j,\ -1-3j$

Answer: not available

GATE 2009 | Q.27

Q.27 An AC source of RMS voltage 20 V with internal impedance $Z_s = (1+2j)\,\Omega$ feeds a load of impedance $Z_L = (7+4j)\,\Omega$ in the figure below. The reactive power consumed by the load is



(A) 8 VAR

(B) 16 VAR

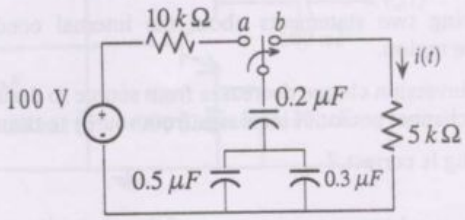
(C) 28 VAR

(D) 32 VAR

Answer: not available

GATE 2009 | Q.28

Q.28 The switch in the circuit shown was on position a for a long time, and is moved to position b at time $t = 0$. The current $i(t)$ for $t > 0$ is given by



(A) $0.2e^{-125t}u(t)$ mA

(B) $20e^{-1250t}u(t)$ mA

(C) $0.2e^{-1250t}u(t)$ mA

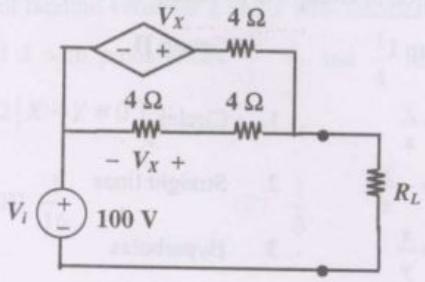
(D) $20e^{-1000t}u(t)$ mA

Q.29 In the circuit shown, what value of R maximizes the power delivered to R?

Answer: not available

GATE 2009 | Q.29

Q.29 In the circuit shown, what value of R_L maximizes the power delivered to R_L?



(A) 2.4 Ω (B) $\frac{8}{3}\Omega$ (C) 4 Ω (D) 6 Ω

Answer: not available

GATE 2009 | Q.30

Q.30 The time domain behavior of an RL circuit is represented by

$$L\frac{di}{dt}+Ri=V_o\left(1+Be^{-Rt/L}\sin t\right)u(t).$$

For an initial current of $i(0)=\frac{V_o}{R}$, the steady state value of the current is given by

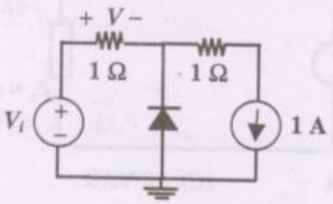
(A) $i(t)\to\frac{V_0}{R}$ (B) $i(t)\to\frac{2V_0}{R}$

(C) $i(t)\to\frac{V_0}{R}(1+B)$ (D) $i(t)\to\frac{2V_0}{R}(1+B)$

Answer: not available

GATE 2009 | Q.31

Q.31 In the circuit below, the diode is ideal. The voltage V is given by



(A) $\min(V_i,1)$ (B) $\max(V_i,1)$ (C) $\min(-V_i,1)$ (D) $\max(-V_i,1)$

Answer: not available

GATE 2009 | Q.32

Q.32 Consider the following two statements about the internal conditions in an n-channel MOSFET operating in the active region.

S1: The inversion charge decreases from source to drain
S2: The channel potential increases from source to drain

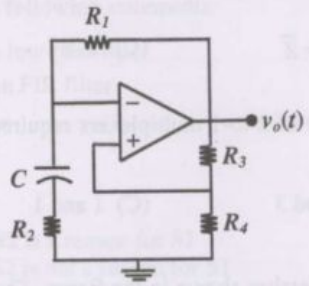
Which of the following is correct ?

(A) Only S2 is true
(B) Both S1 and S2 are false.
(C) Both S1 and S2 are true, but S2 is not a reason for S1
(D) Both S1 and S2 are true, and S2 is a reason for S1

Answer: not available

GATE 2009 | Q.33

Q.33 In the following astable multivibrator circuit, which properties of $v_o(t)$ depend on R_2 ?

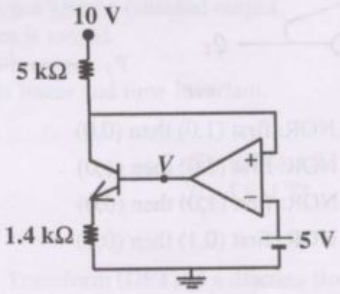


(A) Only the frequency
(B) Only the amplitude
(C) Both the amplitude and the frequency
(D) Neither the amplitude nor the frequency

Answer: not available

GATE 2009 | Q.34

Q.34 In the circuit shown below, the op-amp is ideal, the transistor has $V_{BE} = 0.6$ V and $\beta = 150$. Decide whether the feedback in the circuit is positive or negative and determine the voltage V at the output of the op-amp.

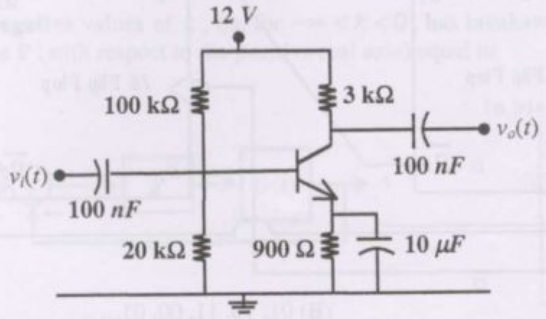


(A) Positive feedback, $V = 10$ V
(B) Positive feedback, $V = 0$ V
(C) Negative feedback, $V = 5$ V
(D) Negative feedback, $V = 2$ V

Answer: not available

GATE 2009 | Q.35

Q.35 A small signal source $v_i(t) = A\cos 20t + B\sin 10^6 t$ is applied to a transistor amplifier as shown below. The transistor has $\beta = 150$ and $h_{ie} = 3\,\text{k}\Omega$. Which expression best approximates $v_o(t)$?



(A) $v_o(t) = -1500\left(A\cos 20t + B\sin 10^6 t\right)$ (B) $v_o(t) = -150\left(A\cos 20t + B\sin 10^6 t\right)$

(C) $v_o(t) = -1500B\sin 10^6 t$ (D) $v_o(t) = -150B\sin 10^6 t$

EC

Answer: not available

GATE 2009 | Q.36

Q.36 If $X = 1$ in the logic equation $\left[X + Z\left\{\overline{Y} + \left(\overline{Z} + X\overline{Y}\right)\right\}\right]\left\{\overline{X} + \overline{Z}\left(X + Y\right)\right\} = 1$, then

(A) $Y = Z$ (B) $Y = \overline{Z}$ (C) $Z = 1$ (D) $Z = 0$

Answer: not available

GATE 2009 | Q.37

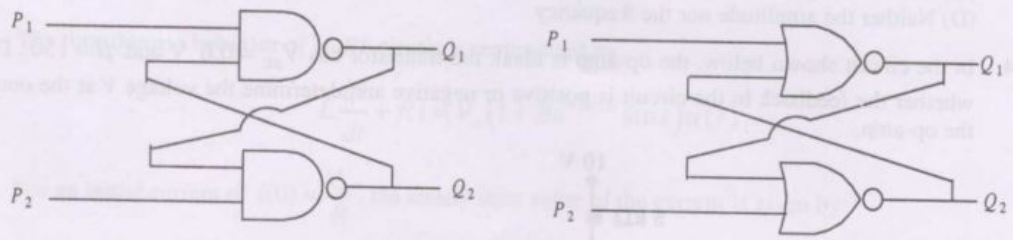
Q.37 What are the minimum number of 2-to-1 multiplexers required to generate a 2-input AND gate and a 2-input Ex-OR gate ?

(A) 1 and 2 (B) 1 and 3 (C) 1 and 1 (D) 2 and 2

Answer: not available

GATE 2009 | Q.38

Q.38 Refer to the NAND and NOR latches shown in the figure. The inputs (P_1, P_2) for both the latches are first made $(0,1)$ and then, after a few seconds, made $(1,1)$. The corresponding stable outputs (Q_1, Q_2) are



(A) NAND: first (0,1) then (0,1) NOR: first (1,0) then (0,0)

(B) NAND: first (1,0) then (1,0) NOR: first (1,0) then (1,0)

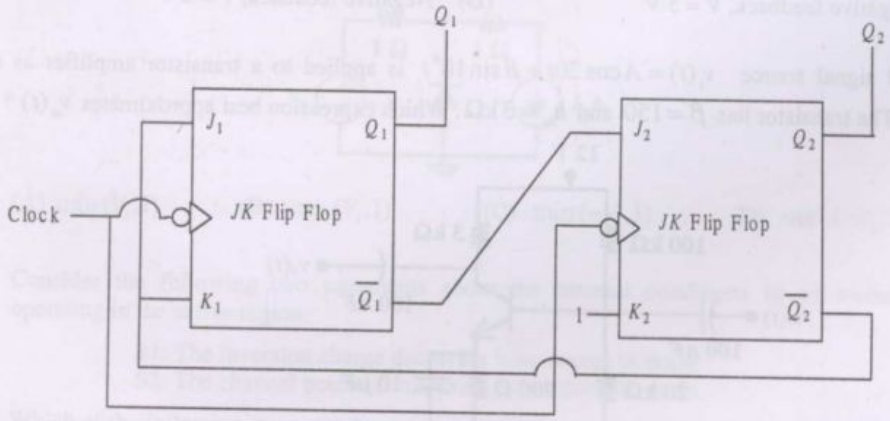
(C) NAND: first (1,0) then (1,0) NOR: first (1,0) then (0,0)

(D) NAND: first (1,0) then (1,1) NOR: first (0,1) then (0,1)

Answer: not available

GATE 2009 | Q.39

Q.39 What are the counting states (Q_1, Q_2) for the counter shown in the figure below ?



(A) 11, 10, 00, 11, 10,....
(B) 01, 10, 11, 00, 01,....
(C) 00, 11, 01, 10, 00,....
(D) 01, 10, 00, 01, 10,....

Answer: not available

GATE 2009 | Q.40

Q.40 A system with transfer function $H(z)$ has impulse response $h(\cdot)$ defined as $h(2)=1$, $h(3)=-1$ and $h(k)=0$ otherwise. Consider the following statements.

S1: $H(z)$ is a low-pass filter.
S2: $H(z)$ is an FIR filter.

Which of the following is correct ?

(A) Only S2 is true
(B) Both S1 and S2 are false
(C) Both S1 and S2 are true, and S2 is a reason for S1
(D) Both S1 and S2 are true, but S2 is not a reason for S1

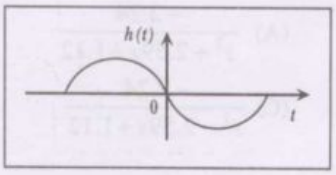
Answer: not available

GATE 2009 | Q.41

Q.41 Consider a system whose input x and output y are related by the equation

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(2\tau)d\tau,$$

where $h(t)$ is shown in the graph.



Which of the following four properties are possessed by the system ?
BIBO: Bounded input gives a bounded output.
Causal: The system is causal.
LP: The system is low pass.
LTI: The system is linear and time-invariant.

(A) Causal, LP
(B) BIBO, LTI
(C) BIBO, Causal, LTI
(D) LP, LTI

Answer: not available

GATE 2009 | Q.42

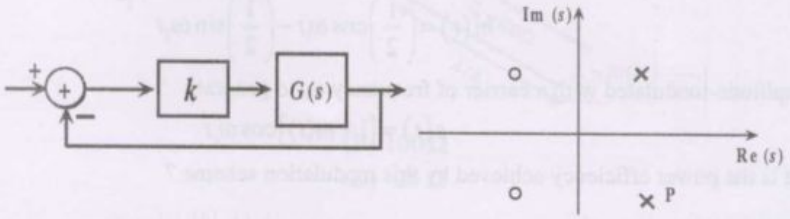
Q.42 The 4-point Discrete Fourier Transform (DFT) of a discrete time sequence {1, 0, 2, 3} is

(A) [0, –2 + 2j, 2, –2 –2j]
(B) [2, 2 + 2j, 6, 2 – 2j]
(C) [6, 1 – 3j, 2, 1 + 3j]
(D) [6, –1 + 3j, 0, –1 –3j]

Answer: not available

GATE 2009 | Q.43

Q.43 The feedback configuration and the pole-zero locations of $G(s)=\frac{s^2-2s+2}{s^2+2s+2}$ are shown below. The root locus for **negative** values of k, i.e. for $-\infty < k < 0$, has breakaway/ break-in points and angle of departure at pole **P** (with respect to the positive real axis) equal to



(A) $\pm\sqrt{2}$ and 0°
(B) $\pm\sqrt{2}$ and 45°
(C) $\pm\sqrt{3}$ and 0°
(D) $\pm\sqrt{3}$ and 45°

Answer: not available

GATE 2009 | Q.44

Q.44 An LTI system having transfer function $\frac{s^2+1}{s^2+2s+1}$ and input $x(t) = \sin(t+1)$ is in steady state. The output is sampled at a rate ω_s rad/s to obtain the final output $\{y(k)\}$. Which of the following is true ?

(A) $y(\cdot)$ is zero for all sampling frequencies ω_s
(B) $y(\cdot)$ is nonzero for all sampling frequencies ω_s
(C) $y(\cdot)$ is nonzero for $\omega_s > 2$, but zero for $\omega_s < 2$
(D) $y(\cdot)$ is zero for $\omega_s > 2$, but nonzero for $\omega_s < 2$

Answer: not available

GATE 2009 | Q.45

Q.45 The unit step response of an under-damped second order system has steady state value of –2. Which one of the following transfer functions has these properties ?

(A) $\frac{-2.24}{s^2+2.59s+1.12}$
(B) $\frac{-3.82}{s^2+1.91s+1.91}$
(C) $\frac{-2.24}{s^2-2.59s+1.12}$
(D) $\frac{-3.82}{s^2-1.91s+1.91}$

Answer: not available

GATE 2009 | Q.46

Q.46 A discrete random variable X takes values from 1 to 5 with probabilities as shown in the table. A student calculates the mean of X as 3.5 and her teacher calculates the variance of X as 1.5. Which of the following statements is true ?

k	1	2	3	4	5
$P(X=k)$	0.1	0.2	0.4	0.2	0.1

(A) Both the student and the teacher are right
(B) Both the student and the teacher are wrong
(C) The student is wrong but the teacher is right
(D) The student is right but the teacher is wrong

Answer: not available

GATE 2009 | Q.47

Q.47 A message signal given by

$$m(t)=\left(\frac{1}{2}\right)\cos\omega_1 t-\left(\frac{1}{2}\right)\sin\omega_2 t$$

is amplitude-modulated with a carrier of frequency ω_c to generate

$$s(t)=[1+m(t)]\cos\omega_c t$$

What is the power efficiency achieved by this modulation scheme ?

(A) 8.33 %
(B) 11.11 %
(C) 20 %
(D) 25 %

EC

Q.48 A communication channel with AWGN operating at a signal to noise ratio $SNR \gg 1$ and bandwidth B

Answer: not available

GATE 2009 | Q.48

Q.48 A communication channel with AWGN operating at a signal to noise ratio $SNR \gg 1$ and bandwidth B has capacity C_1. If the SNR is doubled keeping B constant, the resulting capacity C_2 is given by

(A) $C_2 \approx 2C_1$
(B) $C_2 \approx C_1 + B$
(C) $C_2 \approx C_1 + 2B$
(D) $C_2 \approx C_1 + 0.3B$

Answer: not available

GATE 2009 | Q.49

Q.49 A magnetic field in air is measured to be

$$\vec{B} = B_0\left(\frac{x}{x^2+y^2}\hat{y} - \frac{y}{x^2+y^2}\hat{x}\right)$$

What current distribution leads to this field ? [**Hint**: The algebra is trivial in cylindrical coordinates.]

(A) $\vec{J} = -\frac{B_0\hat{z}}{\mu_0}\left(\frac{1}{x^2+y^2}\right),\ r \neq 0$

(B) $\vec{J} = -\frac{B_0\hat{z}}{\mu_0}\left(\frac{2}{x^2+y^2}\right),\ r \neq 0$

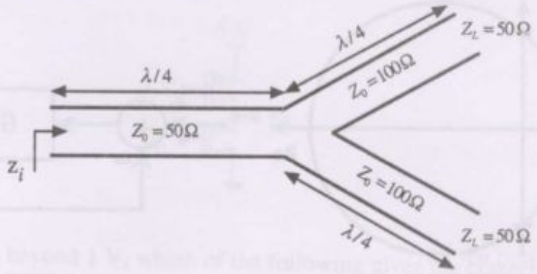
(C) $\vec{J} = 0,\ r \neq 0$

(D) $\vec{J} = \frac{B_0\hat{z}}{\mu_0}\left(\frac{1}{x^2+y^2}\right),\ r \neq 0$

Answer: not available

GATE 2009 | Q.50

Q.50 A transmission line terminates in two branches, each of length $\lambda/4$, as shown. The branches are terminated by $50\,\Omega$ loads. The lines are lossless and have the characteristic impedances shown. Determine the impedance Z_i as seen by the source.



(A) $200\,\Omega$

(B) $100\,\Omega$

(C) $50\,\Omega$

(D) $25\,\Omega$

Answer: not available

GATE 2009 | Q.51

Q.51 The built-in potential of the junction

(A) is 0.70 V

(B) is 0.76 V

(C) is 0.82 V

(D) cannot be estimated from the data given

Answer: not available

GATE 2009 | Q.52

Q.52 The peak electric field in the device is

(A) 0.15 $MV \cdot cm^{-1}$, directed from p-region to n-region
(B) 0.15 $MV \cdot cm^{-1}$, directed from n-region to p-region
(C) 1.80 $MV \cdot cm^{-1}$, directed from p-region to n-region
(D) 1.80 $MV \cdot cm^{-1}$, directed from n-region to p-region

Answer: not available

GATE 2009 | Q.53

Q.53 Which of the following statements is true ?

(A) $G(s)$ is an all-pass filter
(B) $G(s)$ has a zero in the right-half plane
(C) $G(s)$ is the impedance of a passive network
(D) $G(s)$ is marginally stable

Q.54 The gain and phase margins of $G(s)$ for closed loop stability are

Answer: not available

GATE 2009 | Q.54

Q.54 The gain and phase margins of $G(s)$ for closed loop stability are

(A) 6 dB and 180° (B) 3 dB and 180° (C) 6 dB and 90° (D) 3 dB and 90°

Answer: not available

GATE 2009 | Q.55

Q.55 If the signal to quantization noise ratio required in uniformly quantizing the signal is 43.5 dB, the step size of the quantization is approximately

(A) 0.0333 V (B) 0.05 V (C) 0.0667 V (D) 0.10 V

Answer: not available

GATE 2009 | Q.56

Q.56 If the positive values of the signal are uniformly quantized with a step size of 0.05 V, and the negative values are uniformly quantized with a step size of 0.1 V, the resulting signal to quantization noise ratio is approximately

(A) 46 dB (B) 43.8 dB (C) 42 dB (D) 40 dB

Answer: not available

GATE 2009 | Q.57

Q.57 For small increase in V_G beyond 1 V, which of the following gives the correct description of the region of operation of each MOSFET ?

(A) Both the MOSFETs are in saturation region
(B) Both the MOSFETs are in triode region
(C) n-MOSFET is in triode and p-MOSFET is in saturation region
(D) n-MOSFET is in saturation and p-MOSFET is in triode region

EC

... for V_G = 1.5 V. [**Hint:** Use the appropriate current-voltage equation for

Answer: not available

GATE 2009 | Q.58

Q.58 Estimate the output voltage V_0 for V_G = 1.5 V. [**Hint:** Use the appropriate current-voltage equation for each MOSFET, based on the answer to Q.57.]

(A) $4-\frac{1}{\sqrt{2}}$ V (B) $4+\frac{1}{\sqrt{2}}$ V (C) $4-\frac{\sqrt{3}}{2}$ V (D) $4+\frac{\sqrt{3}}{2}$ V

Answer: not available

GATE 2009 | Q.59

Q.59 If segments a to g are considered as functions of P_1 and P_2, then which of the following is correct ?

(A) $g=\overline{P}_1+P_2,\ d=c+e$
(B) $g=P_1+P_2,\ d=c+e$
(C) $g=\overline{P}_1+P_2,\ e=b+c$
(D) $g=P_1+P_2,\ e=b+c$

Answer: not available

GATE 2009 | Q.60

Q.60 What are the minimum numbers of NOT gates and 2-input OR gates required to design the logic of the driver for this 7-segment display ?

(A) 3 NOT and 4 OR
(B) 2 NOT and 4 OR
(C) 1 NOT and 3 OR
(D) 2 NOT and 3 OR

END OF THE QUESTION PAPER

Answer: not available